

are authentic. Automakers can apply these protections through secure firmware management, over-the-air software updates, and open source transparency.

To ensure cybersecurity for the connected car, a comprehensive approach is needed, from monitoring software components to deploying secure remote updates and leveraging open collaboration communities to find and resolve faults earlier. Automakers need to understand that cybersecurity cannot be added later.

In addition to the road, AI has transformed manufacturing, logistics, and agriculture. Smart cameras will monitor the factory production line in real time to minimise human error and increase quality. Industrial laundries use AI vision systems to help sort and detect defects, while warehouses will make use of fleets of AI-driven robots to coordinate logistics and speed up delivery, a strategy perfected by leaders such as Amazon and Flipkart.

Now, in the fields of autonomous tractors, drone surveillance, and precision irrigation systems, edge AI and TinyML have optimised resource utilisation to the last step. Such intelligent and decentralised systems make immediate, secure decisions, thereby saving time and energy.

Intelligence at the edge

Edge AI represents a fundamental shift in embedded systems, moving intelligence from centralised cloud servers directly onto devices. This transition delivers significant advantages. Local data processing reduces latency, enabling real-time responses essential for mission-critical applications. It also improves cost efficiency by reducing dependence on cloud infrastructure and lowering operational expenses.

Also, edge AI reinforces autonomy, enabling devices to function seamlessly without

Embedded tech trends outline

The trends reflect a clear direction: faster, safer, and smarter systems that think for themselves yet remain deeply connected to human needs. The next decade will belong to designs that fuse computing power with creativity, building a world where embedded systems don't just power devices but drive innovation itself.

Trend	Impact
AI integration	On-device intelligence and predictive analytics
Edge computing	Low-latency, privacy-first data processing
5G and RedCap	Reliable connectivity for industrial IoT
Power optimisation	Smarter chips, longer battery life
Hardware security	Secure boot, encryption, SBOMs
Open source adoption	Transparent, faster innovation
AI-driven development	Generative tools assist engineers
Smart manufacturing	Robotics and automation powered by AI

constant internet access. For instance, smart home systems can manage lights, fans, and appliances locally without relying on the cloud, while complex tasks related to natural language processing might still use the cloud. Developers will be working on a new class of hardware-optimised algorithms, TinyML solutions, and localised approaches to deploy AI.

Open source dominance

Open source software plays a central role in embedded systems development. Platforms such as Zephyr, FreeRTOS, and Linux provide reliable and accessible tools that accelerate device development. Pre-built components reduce development time, community-driven validation helps identify issues quickly, and collaborative ecosystems simplify the integration of advanced capabilities such as AI, connectivity, and device communication.

A significant proportion of embedded devices now rely on open source software, which continues to shape industry standards and drive innovation.

Generative AI is changing how this software is made. It can write simple code, improve programs, and run automated tests, so engineers can spend more time designing systems, linking features, and thinking of new ideas. This also changes jobs; engineers now act as system planners, AI guides, and

connection experts. Even with AI, people do need to double-check its work using tests and tools.

Going into 2026, embedded systems are no longer merely background technology; they are increasingly becoming an intelligent, secure, and connected backbone for modern innovation. What once were simple controllers have transformed into AI-powered edge brains that make real-time decisions and operate autonomously across industries.

From wearable health devices that remotely protect lives to self-healing smart factories operating with unmatched efficiency, embedded AI is redefining what's possible. Artificial intelligence, ultra-fast 5G networks, and energy-efficient hardware are improving performance but also driving the need for built-in security.

The development is accelerating with open source platforms and generative AI tools that will help engineers design smarter, safer systems in less time than ever, but the real catch lies in how humans and machines work together. But it is we who guide machine intelligence to develop technology that is not only advanced but also responsible and humane. **EFY**

This article is based on a tech talk session at IEW 2025, Bengaluru, by Mubeen Jukaku, Technology Head, Emertxe Information Technologies Pvt Ltd. It was transcribed and curated by Akanksha Sondhi Gaur, Sr Technical Journalist at EFY.